INCH-POUND
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PERFORMANCE SPECIFICATION

PUMP ASSEMBLY, BILGE, ELECTRIC MOTOR DRIVEN, DIRECT CURRENT

This specification is approved for use by U.S. Army Tank-automotive and Armaments Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the performance and design requirements for an electric motor driven pump assembly which evacuates water from M60 series tanks.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirement documents cited in sections 3 and 4 of this specification, whether or not they are listed.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Tank-automotive and Armaments Command, ATTN: AMSTA-TR-E/BLUE, Warren, MI 48397-5000, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

AMSC N/A FSC 4320

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2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

STANDARDS

DEPARTMENT OF DEFENSE

MIL-STD-462	-	Electromagnetic Interference Characteristics Measurements of.
MIL-STD-810	-	Environmental Test Methods and Engineering Guidelines.

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

ARMY

7059833	-	Bilge Pump Assembly (Interface).

(Copies of these drawings are available from the U.S. Army Tank-automotive and Armaments Command, ATTN: AMSTA-TR-E/BLUE, Warren, MI 48397-5000.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

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AMERICAN NATIONAL STANDARD INSTITUTE (ANSI)

ANSI/ASQC Z1.4 - Sampling Procedures and Tables for Inspection by Attributes (DoD Adopted).

(Application for copies should be addressed to American National Standard Institute (ANSI), 11 West 42nd Street, New York, NY 10036.)

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

MG1-4.21 - Method of Measuring the Motor Vibration (Dynamic Balance) (DoD Adopted).

(Application for copies should be addressed to the National Electrical Manufacturers Association, 2101 L Street, NW, Washington, DC 20037.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified, a sample shall be subjected to first article inspection in accordance with 4.3 (see 6.2).

3.2 Materials. Materials shall be in accordance with the manufacturer's materials specifications for pumps. Materials shall be capable of meeting all operational and environmental requirements specified herein and shall be free of defects which adversely affect performance or serviceability of the finished product. Asbestos and cadmium materials shall not be used in any form in any part of the pump. No item, part or assembly shall contain radioactive materials in which the specific activity is greater than 0.002 microcuries per gram or activity per item equals or exceeds 0.01 microcuries (see 4.5.1).

3.2.1 Metals and finishes. Exposed functional parts shall be fabricated from suitable corrosion resistant material or treated to prevent corrosion. Exposed nonfunctional parts shall be painted white in accordance with the manufacturer's standard (see 4.5.1).

3.2.2 Fungistatic materials. All non-metallic materials shall be inherently fungistatic or treated to resist fungus growth except when used for components of potted or otherwise moisture sealed assemblies (see 4.5.1).

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3.2.3 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided that the materials meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.3 Design and construction. Pump assemblies shall conform to the envelope dimension, inlet/outlet port locations and dimensions, and the electrical and mounting interfaces as shown in drawing 7059833 (see 4.5.1).

3.3.1 Interchangeability. All similar parts including repair parts of the motor assembly furnished on the same order or built to the same drawings shall be completely interchangeable without further machining selective assembly or hand fitting of any kind (see 4.5.1).

3.4 Performance. Unless otherwise specified, the pump assembly shall meet the performance requirements under the following conditions:

a.	Air temperature	73 $\pm$ 18 degrees Fahrenheit (°F)
b.	Pressure	28.5 (+2.0, -4.5) inches of mercury (Hg)
c.	Humidity	50 $\pm$ 30 percent (%)
d.	Voltage	27.5 $\pm$ 0.5 volts (V) direct current (dc)

3.4.1 Voltage range. The pump assembly shall operate between 18 and 30 Vdc. The output at 18 Vdc shall be not less than 30 gallons per minute (gal/min) minimum at 12 foot (ft) total discharge head (TDH) and at 30 Vdc not less than 115 gal/min at a 12 ft TDH (see 4.5.2).

3.4.2 Current draw. The current draw of the pump assembly shall be as shown in figure 1 (see 4.5.3).

3.4.3 Efficiency. The pump assembly shall have an efficiency as shown in figure 1 (see 4.5.4).

3.4.4 Flow rate. The pump assembly shall have a free flow rate of 130 gal/min, a flow rate of 125 gal/min at a 6 ft TDH, and a 30 ft minimum stall TDH (see figure 1 and 4.5.5).

3.4.5 Durability. The pump assembly shall show no evidence of performance degradation after 140 hours (hr) of operation at variable voltage according to the schedule of figure 2, followed by 3600 cycles of operation at one minute off and one minute pumping against a 6 ft TDH with 27 $\pm$ 5 Vdc supplied (see 4.5.6).

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3.4.6 Electromagnetic interference. The assembly shall meet the requirements of figures 3, 4, 5, and 6 for conducted emissions and radiated emissions for ground vehicles. Test equipment shall be located at 1 meter from the item being tested for radiated emissions (see 4.5.7).

3.4.7 Dielectric strength. There shall be no continuing sparkover, no breakdown of dielectric material and no damage to the insulation which would cause failure of the pump assembly when tested in accordance with 4.5.8.

3.4.8 Insulation resistance. The insulation resistance of the pump assembly shall not be less than 12 megohms when tested at 500 Vdc (see 4.5.9).

3.4.9 Dynamic balance. The pump assembly shall conform to the requirements of NEMA Standards Section MG1-4.21. The maximum peak-to-peak displacement in any direction shall not exceed .002 inch (in.) (see 4.5.10).

3.4.10 Temperature. The assembly shall meet the performance requirements specified herein at temperatures ranging from 0°F to +125°F. The assembly shall not be damaged by temperatures ranging from -65°F to +160°F while not operating (see 4.5.11).

3.4.11 Vibration resistance. The assembly shall be capable of operating as specified herein after withstanding the vibration profile per figure 7 for a period of 80 minutes (including up to 4 resonance dwells of 13-1/3 min each) in each of the 3 mutually perpendicular axes, X, Y, and Z, shown in figure 8 (see 4.5.12).

3.4.12 Shock resistance (non-operating). The assembly shall be capable of operating as specified herein after exposure to three half sine wave shock pulses of 40 $\pm$ 4 gravity units (g) peak for a duration of 18 $\pm$ 3 milliseconds (ms) applied in both directions along the 3 mutually perpendicular axes, X, Y, and Z shown in figure 8 (see 4.5.13).

3.4.13 Corrosion resistance. The assembly shall be capable of operating as specified herein after exposure to a 5% sodium chloride atomized spray for 96 hrs (see 4.5.14).

3.4.14 Waterproofness. The pump assembly shall perform as specified herein after it is totally submerged and tested for leakage in accordance with 4.5.15.

3.5 Marking. A nameplate shall be attached securely to the pump assembly as shown on Drawing 7059833. The nameplate shall be marked to include the Military part number, National stock number (NSN), manufacturer's name, serial number and date of manufacture. In addition, the Military part number shall be permanently inscribed on the pump assembly in a clearly visible location not covered by the nameplate (see 4.4.2).

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3.6 Workmanship. The pump assembly shall be free of any electrical, mechanical, or physical defect which would affect its reliability or performance (see 4.4.2).

4. VERIFICATION

4.1 Classification of inspection:

a. First article inspection (see 4.3).
b. Conformance inspections (see 4.4).
    1. Examination (see 4.4.2).
    2. Acceptance tests (see 4.4.3.1).
    3. Control tests (see 4.4.3.2).

4.2 Inspection conditions. Unless otherwise specified, all inspections shall be conducted under the following conditions:

a.	Air temperature	73° ± 18°F
b.	Pressure	28.5 (+2.0, - 4.5) inches of Hg
c.	Humidity	50 ± 30%
d.	Voltage	27.5 ± 0.5 Vdc

4.3 First article inspection. First article inspection shall be performed on preproduction or initial production samples as specified herein.

4.3.1 Preproduction inspection. When specified (see 6.2), the preproduction sample shall consist of two pump assemblies. Preproduction inspection samples shall be examined per table I and tested per table II.

4.3.2 Initial production inspection. Unless otherwise specified (see 6.2), the Government shall select two pump assemblies, from the first twenty pump assemblies produced under the production contract for initial production inspection. Initial production units shall be examined per table I and tested per table II.

4.4 Conformance inspections.

4.4.1 Sampling.

4.4.1.1 Lot formation. An inspection lot shall consist of all the pump assemblies of one type and part number, from an identifiable production period, from one manufacturer, submitted at one time for acceptance.

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4.4.1.2 Sampling for examination. Samples for conformance examination shall be selected in accordance with ANSI/ASQC Z1.4.

4.4.2 Conformance examination. Each sample selected in accordance with 4.4.1.2 shall be examined for defects listed in table I.

4.4.3 Conformance tests.

4.4.3.1 Acceptance tests (100%). Each pump assembly shall be subjected to the tests specified in table II.

4.4.3.2 Control tests. Control tests shall be conducted on one pump assembly from each lot of 200 units consecutively produced, except that not more than one test shall be performed in a three month period, nor less than one test in a six month period. The pump assemblies shall be subjected to the tests specified in table II.

4.5 Methods of inspection. Unless otherwise specified the tests listed in table II shall be performed as specified herein.

4.5.1 Materials and construction. To determine conformance to 3.2 through 3.3.1, inspection and material certification records shall be maintained by the contractor. Records shall be subject to review by the Government and shall include date, part, or characteristic identification, inspection results, and disposition of lot (accepted or rejected). Corrective action taken on noted defects shall be subject to approval by the Government.

4.5.2 Voltage range. To determine conformance to 3.4.1, the assembly shall be operated at an input voltage of 18 Vdc against a 12 ft TDH for 15 min. During the operation, an output flow of 30 gal/min shall be verified. Raise the voltage to 30 Vdc, operate the assembly for an additional 15 min and verify that an output flow of not less than 115 gal/min is obtained. The assembly may be immersed in water for cooling after the initial 15 min of operation.

4.5.3 Current. To determine conformance to 3.4.2, the assembly shall be subjected to 27.5 $\pm$ 0.5 Vdc at a 6 ft TDH. This shall be repeated at 20 ft and 30 ft TDH.

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TABLE I. Classification of defects.

Category	Defect	Method of examination
Major:		
101	Dimensions affecting interchangeability not within tolerances (see 3.3).	SIE 1/
102	Metal not corrosion resistant or inadequately corrosion protected (see 3.2.1).	Visual
103	Exposed nonfunctional parts inadequately painted (see 3.2.1).	Visual
104	Fungistatic materials not furnished or properly certified (see 3.2.2).	Visual
Minor:		
201	Dimensions not affecting interchangeability not within tolerances (see 3.3).	SIE
202	Improper marking (see 3.5).	Visual
203	Improper workmanship (see 3.6).	Visual
204	Pump not packaged as specified (see 5.1).	Visual

1/ SIE = Standard Inspection Equipment.

TABLE II. Classification of tests.

Title	Requirement	Test	First article	Conformance	
				Acceptance (100%)	Control
Voltage range	3.4.1	4.5.2	X		
Current draw	3.4.2	4.5.3	X		X
Efficiency	3.4.3	4.5.4	X		X
Flow rate	3.4.4	4.5.5	X		X
Durability	3.4.5	4.5.6	X		X
Electromagnetic interference	3.4.6	4.5.7	X		X
Dielectric strength	3.4.7	4.5.8	X		
Insulation resistance	3.4.8	4.5.9	X		
Dynamic balance	3.4.9	4.5.10	X		X
Temperature resistance	3.4.10	4.5.11	X		X
High		4.5.11.1			
Low		4.5.11.2			
Vibration resistance	3.4.11	4.5.12	X		
Control		4.5.12.1			X
Shock resistance	3.4.12	4.5.13	X		
Control		4.5.13.1			X
Corrosion resistance	3.4.13	4.5.14	X		
Waterproofness	3.4.14	4.5.15		X	

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4.5.4 Efficiency. To determine conformance to 3.4.3, the assembly shall be subjected to 6, 20, and 30 ft TDH.

4.5.5 Flow rate. To determine conformance to 3.4.4, the assembly shall be subjected to 0, 6, and 30 ft TDH.

4.5.6 Durability. To determine conformance to 3.4.5, the pump assembly shall be subjected to 140 hrs of continuous running at each of the voltage variations of figure 2. After the initial 15 min of operation the body of the assembly may be submerged in water for cooling. After 140 hrs of continuous operation, subject the assembly to 3600 cycles on operation consisting of one min pumping a 6 ft TDH at 27.5 $\pm$ 0.5 Vdc followed by one minute at zero voltage. The body of the pump may be submerged in water for cooling. Upon completion of the cycling test subject the assembly to the efficiency test of 4.5.4.

4.5.6.1 Control. The control test shall be the same as 4.5.6 except the continuous running shall be for six hours at each voltage level of figure 2 for a total of 36 hrs, and the cycling shall be for a total of 15 hrs of "on" time.

4.5.7 Electromagnetic interference. To verify conformance to 3.4.6, subject the assembly to the electromagnetic interference tests specified in MIL-STD-462. At the conclusion of this test a certification shall be provided attesting that the assembly meets the corresponding requirements.

4.5.8 Dielectric strength. To verify conformance to 3.4.7, prior to final assembly, perform the dielectric strength test on the armature and field windings separately in accordance with appendix A. Apply 500 volts root means square (Vrms) at a frequency of 60 Hertz (Hz) for a period of 75 $\pm$ 5 seconds. Verify that there is no continuing sparkover, no breakdown of dielectric material, and no damage to the insulation which would cause failure of the pump assembly.

4.5.9 Insulation resistance. Prior to final assembly, to verify conformance to 3.4.8, stabilize the assembly at 77°F $\pm$ 10°F and measure the insulation resistance between (1) the positive lead and frame ground for each field circuit, and (2) between commutator and shaft for the armature circuit in accordance with appendix B. Verify that the resistance is not less than 12 megohms for each field circuit and for the armature circuit including the insulated brush holders.

4.5.10 Dynamic balance. To determine conformance to 3.4.9, place the assembly on an elastic mounting in an operating position. The weight of the assembly should compress an elastic mounting approximately 0.25 in., or position the assembly such that the natural frequency is one quarter or less of the operating revolutions per minute (rpm). Operate the assembly at reduced voltage, full load speed. Measure the amplitude of running vibration at the drive end in three axes and verify that it does not exceed 0.002 in. displacement.

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4.5.11 Temperature resistance. To verify conformance to 3.4.10 perform the tests of 4.5.11.1 and 4.5.11.2.

4.5.11.1 High temperature. To determine conformance to 3.4.10, test the pump assembly at the required high temperature in accordance with MIL-STD-810, method 501.3, procedure II. At the conclusion of step 4 (at 160°F), stabilize the assembly and water to be used in the test at 125°F and test the pump assembly in accordance with 4.5.4. During the initial 15 min of operation, only the pump section of the assembly shall be submerged. Return the pump assembly to ambient temperature and test in accordance with 4.5.4.

4.5.11.2 Low temperature. To determine conformance to 3.4.10, test the pump assembly at the required low temperature in accordance with MIL-STD-810, method 502.3, procedure 1. Maintain the temperature of step 2 at 65°F for a period of 12 hrs minimum. At the conclusion of this time, stabilize the pump assembly at 0°F and test it in accordance with 4.5.4. Return the assembly to standard ambient temperature and test in accordance with 4.5.4.

4.5.12 Vibration resistance. To determine conformance to 3.4.11, the assembly shall be mounted as shown in figure 8 and subjected to the vibration test specified in MIL-STD-810, method 514.4, procedure VIII, except as follows:

a. Test level shall be the vibration curve shown in figure 7. Time schedule shall be 40 min per axis including 6-2/3 min dwell time at each resonance.
b. Connections and instrumentation shall be attached to the assembly to permit testing as specified in c through g.
c. Conduct resonant frequency search along each axis at room temperature and with input power applied over the frequency range of 5 to 500 Hz. Up to four resonant frequencies shall be identified (the four most severe resonant frequencies if more than four are present) in each axis for the resonant dwell test.
d. Place the assembly in a 125°F ambient temperature chamber.
e. Conduct sinusoidal vibration cycling and resonant dwell tests at 125°F and at the test levels and time schedules identified in “a” above.
f. Repeat the test described in “d” and “e” above except the ambient temperature shall be 0°F.
g. At the conclusion of the vibration test of “f” above return the assembly to room temperature and subject it to the efficiency test of 4.5.4.

4.5.12.1 Control. The control test shall be the same as 4.5.12 except the temperature shall be 73 $\pm$ 18°F and the time schedule shall be 40 min per axis including two min dwell time at each resonance.

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4.5.13 Shock resistance. To determine conformance to 3.4.12 mount the assembly as specified in figure 8 and apply three half wave sine pulses in both directions along each of 3 mutually perpendicular axes, X, Y, and Z. Test in accordance with MIL-STD-810, method 516.4, procedure I for a total of 18 shock pulses. Peak amplitude shall be 40 $\pm$ 4 g with a time duration of 18 $\pm$ 3 msec measured at the 10% amplitude points. At the conclusion of this test, test the assembly in accordance with 4.5.4.

4.5.13.1 Control. This test shall be the same as 4.5.13, except that the shock level shall be limited to 30 $\pm$ 3g.

4.5.14 Corrosion resistance. To determine conformance to 3.4.13, subject the assembly to the corrosion resistance test specified in MIL-STD-810, method 509.3. At the conclusion of this test, test the assembly in accordance with 4.5.4.

4.5.15 Waterproofness. To determine conformance to 3.4.14, the assembly shall be submerged in clear water at ambient temperature with the electrical connector capped and with the uppermost surface of the assembly a minimum of one in. below the surface of the water. A pressure differential of 6 pounds per square inch (psi) shall be developed such that the assembly's internal air pressure exceeds the external pressure. The pressure differential shall be maintained for a two min period without evidence of leakage. Bubbles which are the result of entrapped air on exterior surfaces of the assembly shall not be considered a leak. The pump assembly shall not be electrically operated during immersion. At the conclusion of the test, subject the assembly to the efficiency tests of 4.5.4.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature which may be helpful, but is not mandatory.)

6.1 Intended use. The assembly is intended for use on the M60 series tanks to evacuate water from the crew compartment.

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6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Applicable drawing (see 3.3).
c. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2.1 and 2.3).
d. When first article is required (see 3.1).
e. If inspection conditions should be other than as specified (see 4.2).
f. If preproduction inspection is required (see 4.3.1).
g. If initial production inspection is required (see 4.3.2).
h. Packaging requirements (see 5.1).

6.3 Subject term (key word) listing.

Flow rate
Total discharge head
Water evacuation
Waterproofness

6.4 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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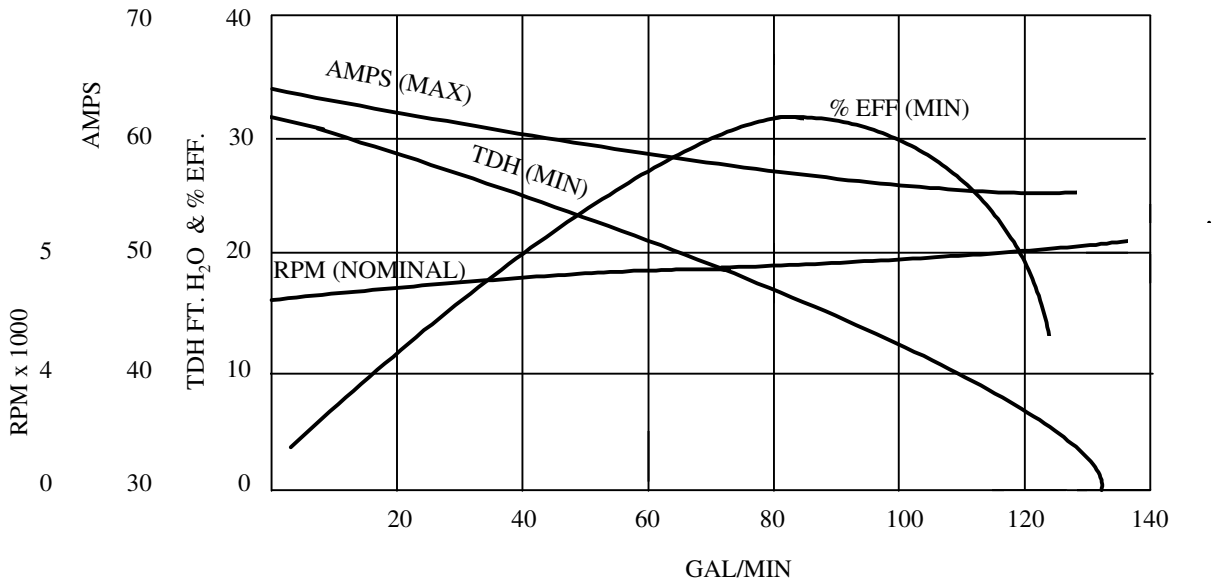


FIGURE 1. Performance curve.

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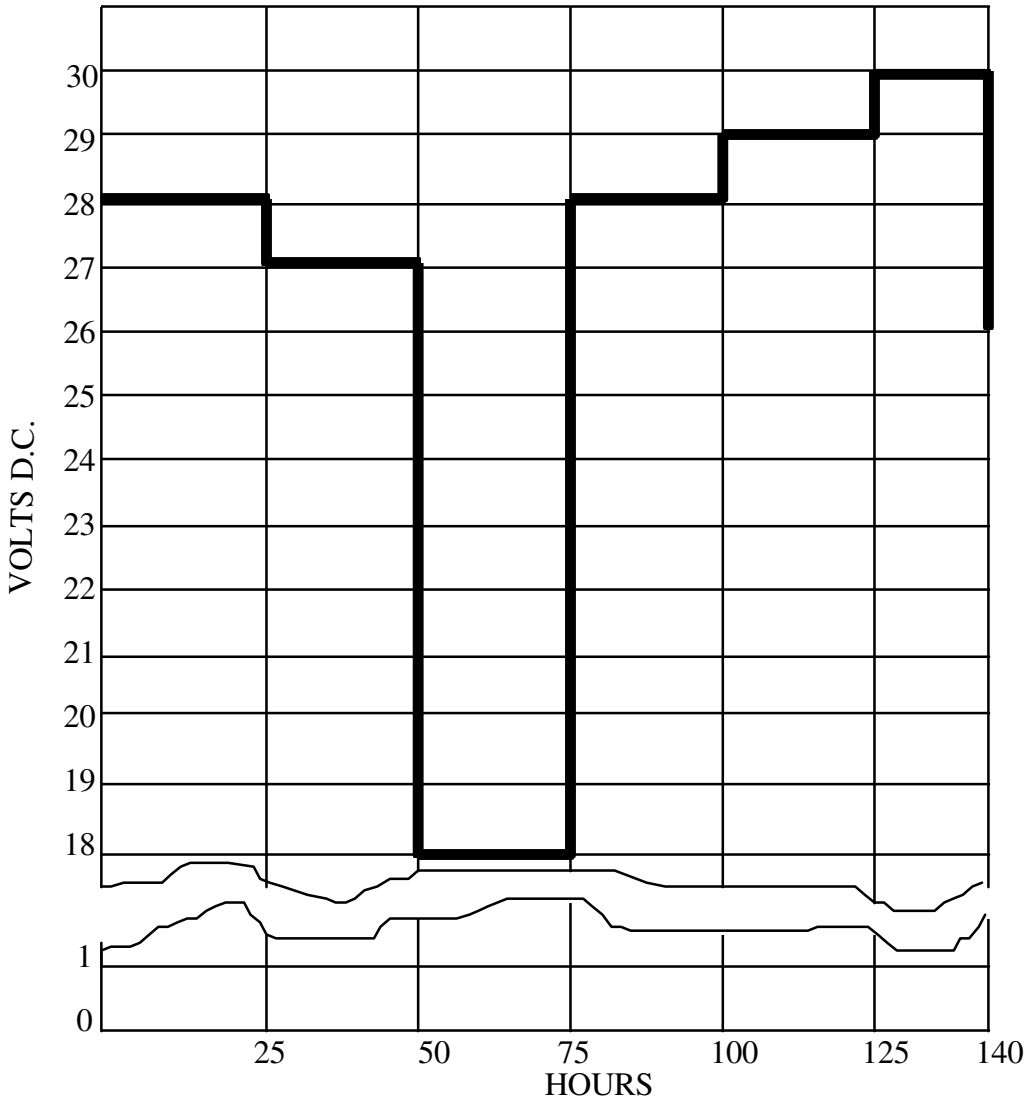


FIGURE 2. Endurance - continuous running.

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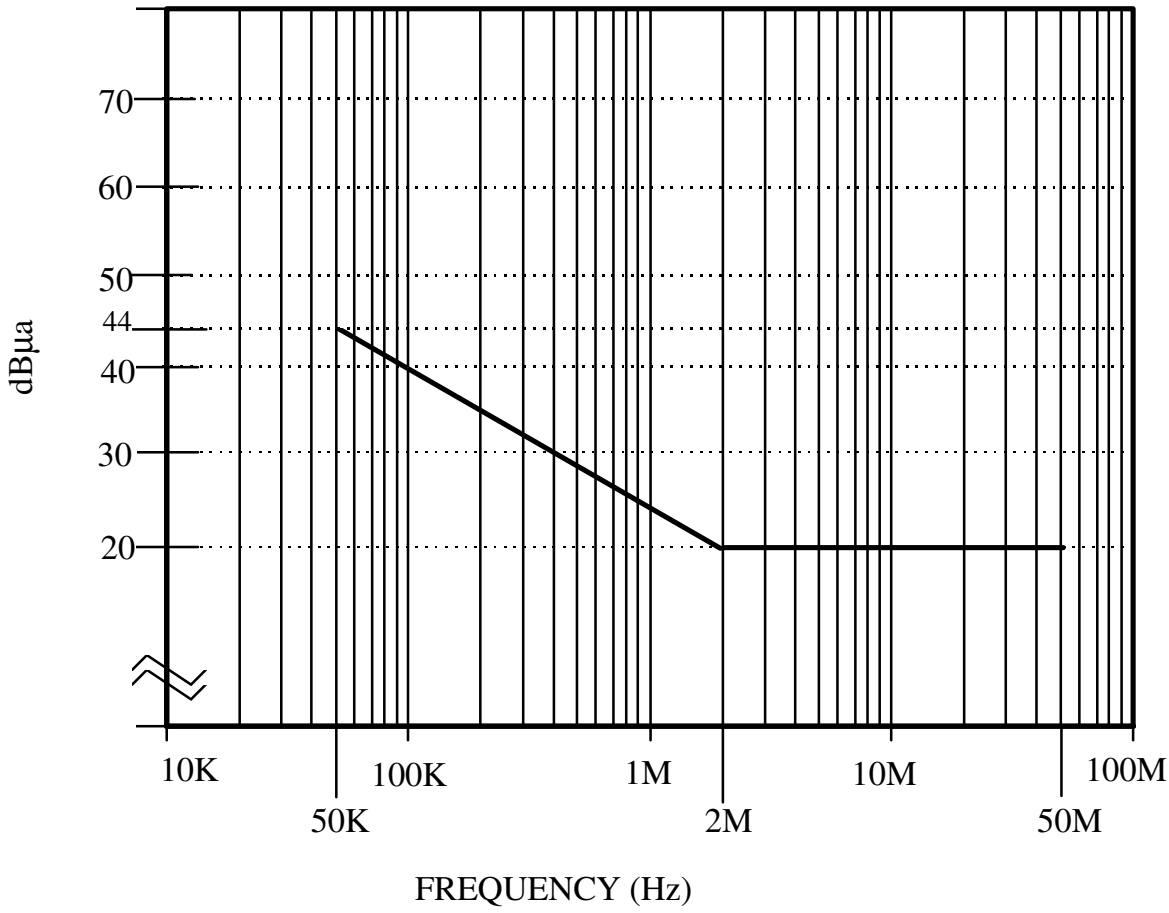


FIGURE 3. Limits for narrowband conducted emissions.

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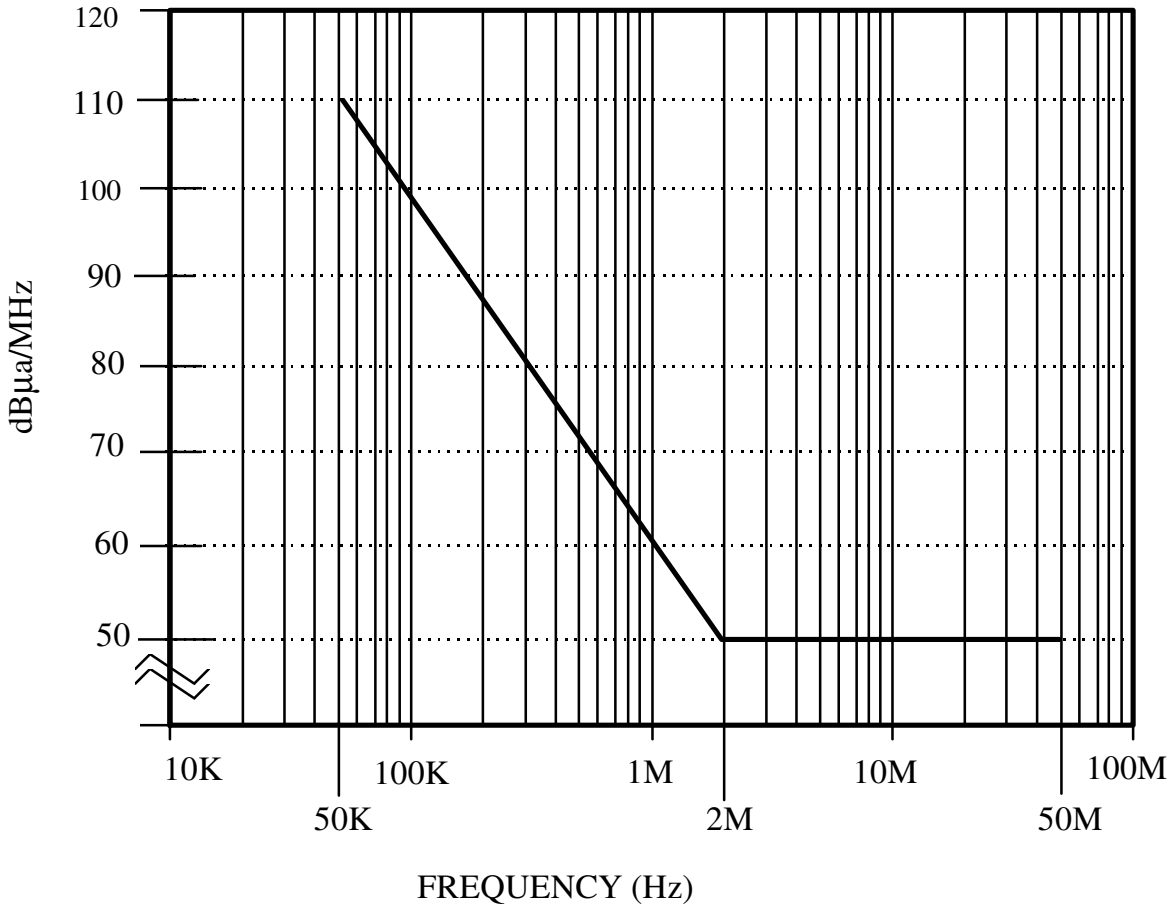


FIGURE 4. Limits for broadbands conducted emissions.

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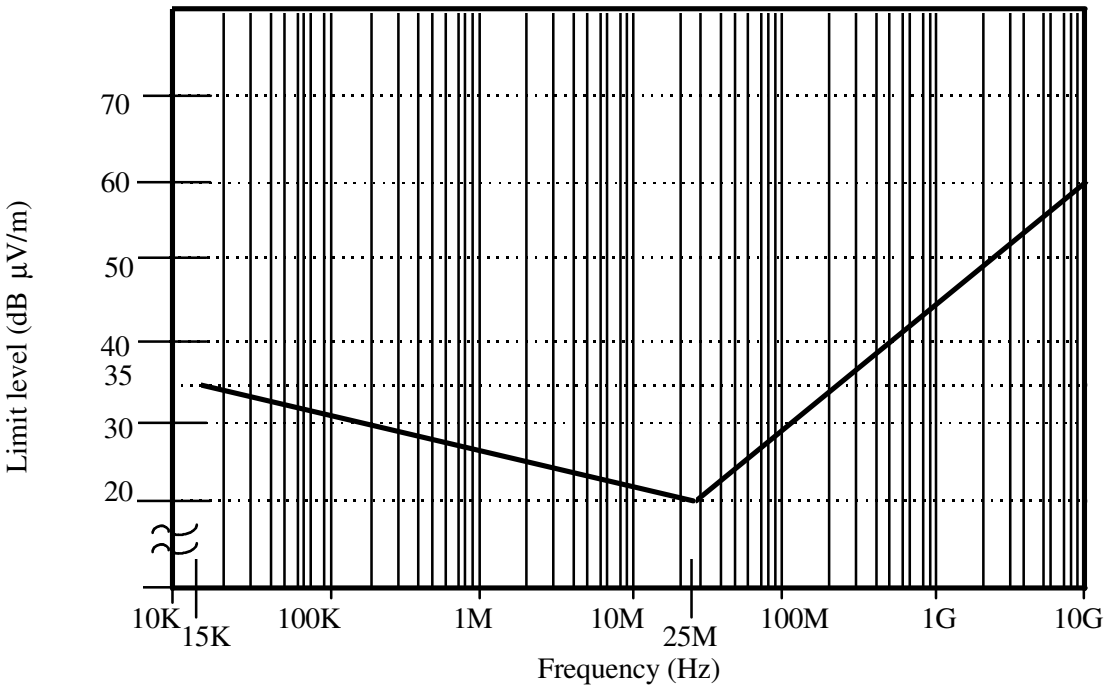


FIGURE 5. Limits for narrowband radiated emissions.

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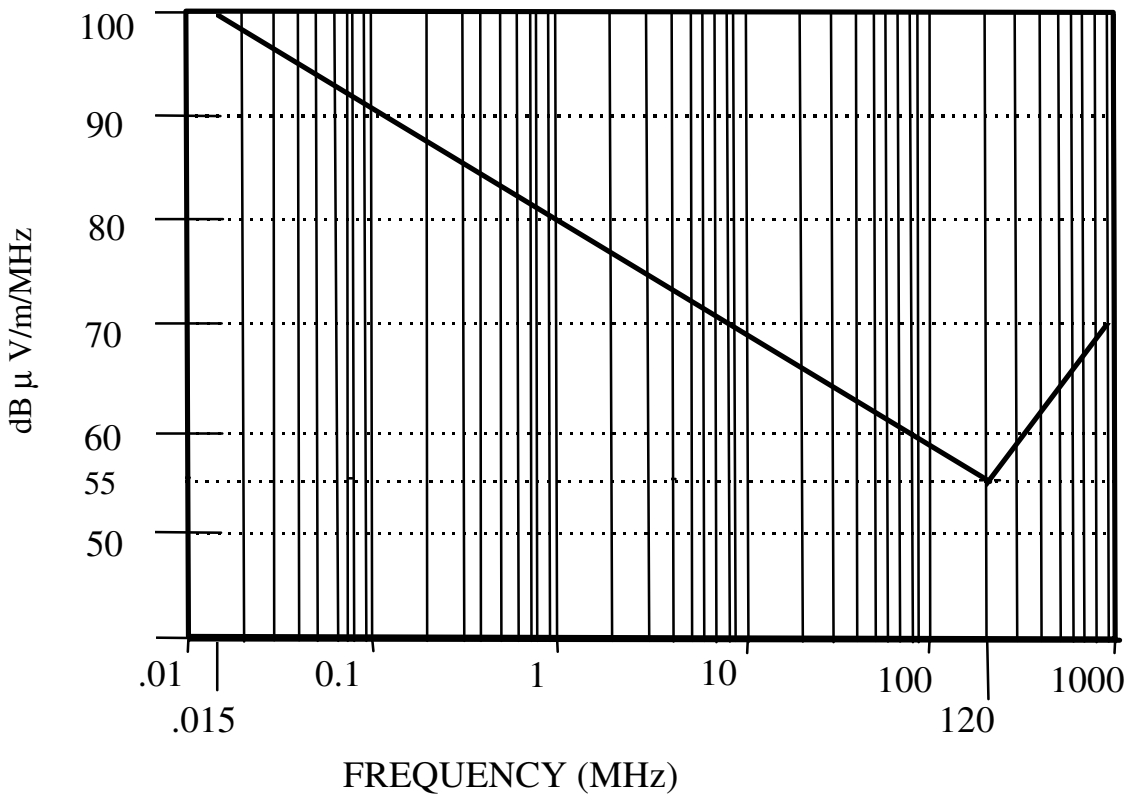


FIGURE 6. Limits for broadband radiated emissions.

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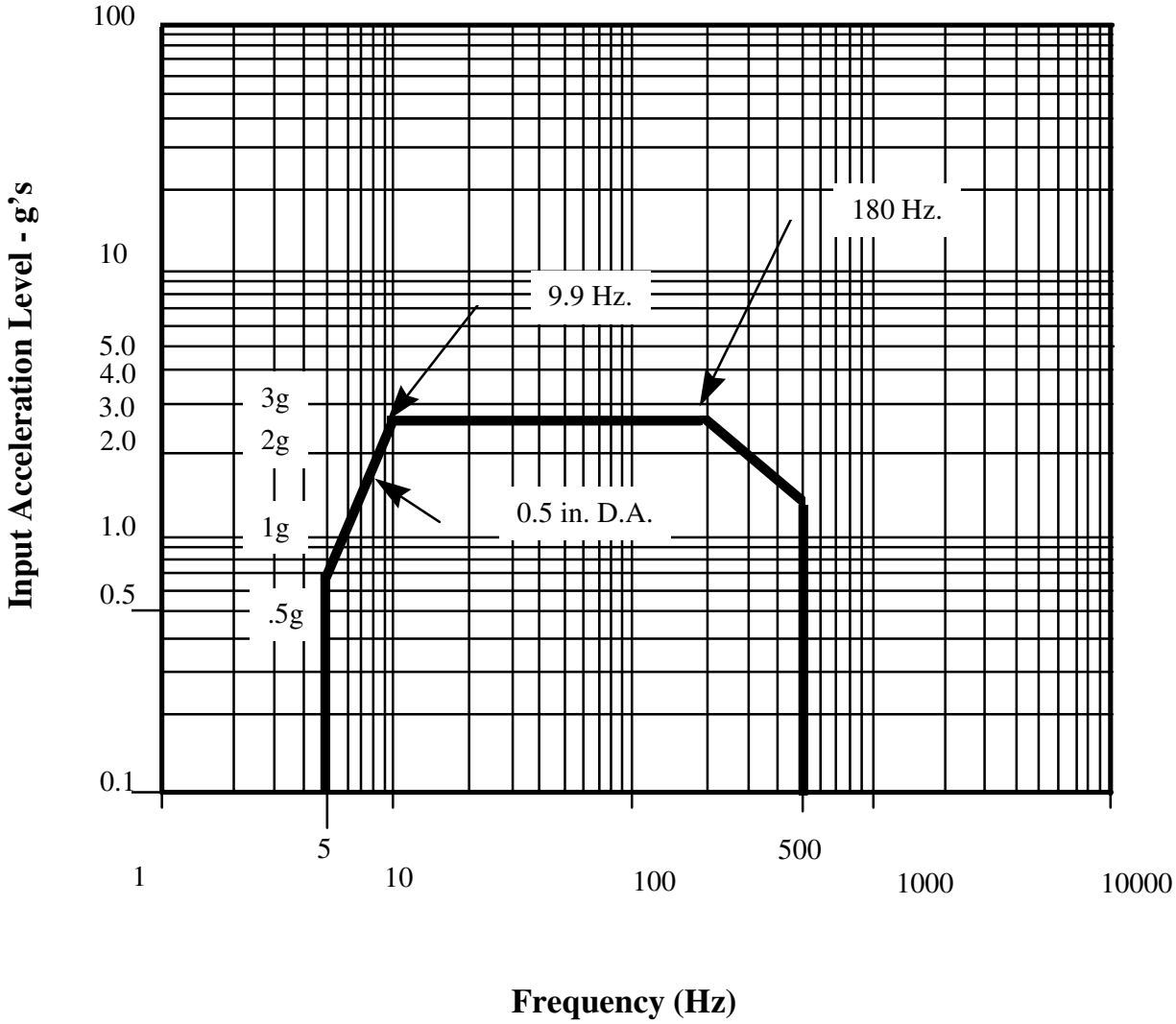


FIGURE 7. Vibration profile.

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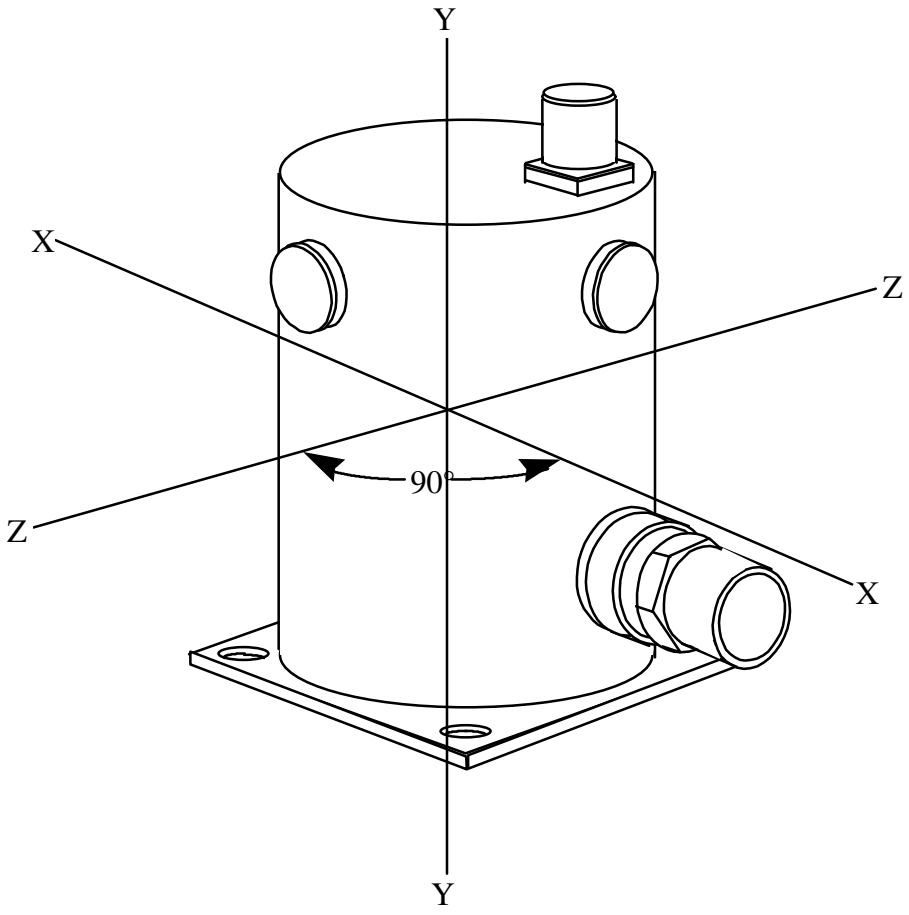


FIGURE 8. Shock and vibration axis.

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APPENDIX A

DIELECTRIC WITHSTANDING VOLTAGE

A.1 SCOPE

A.1.1 Scope. The appendix details the procedure for the dielectric withstanding voltage test (also called high-potential, over potential, voltage-breakdown, or dielectric-strength test) consists of the application of a voltage higher than rated voltage for a specific time between mutually insulated portions of a component part or between insulated portions and ground. This is used to prove that the component part can operate safely at its rated voltage and withstand momentary overpotentials due to switching, surges, and other similar phenomena. Although this test is often called a voltage breakdown or dielectric-strength test, it is not intended that this test cause insulation breakdown or that it be used for detecting corona, rather it serves to determine whether insulating materials and spacings in the component part are adequate. When a component part is faulty in these respects, application of the test voltage will result in either disruptive discharge or deterioration. Disruptive, discharge is evidenced by flashover (surface discharge), sparkover (air discharge), or breakdown (puncture discharge). Deterioration due to excessive leakage currents may change electrical parameters or physical characteristics.

A.2 APPLICABLE DOCUMENTS

(This section is not applicable to this appendix.)

A.3 PROCEDURE

A.3.1 Preparation. When special preparations or conditions such as special test fixtures, reconnections, grounding, isolation, or immersion in water are required, they shall be specified.

A.3.2 Test voltage. Specimens shall be subjected to a test voltage of the magnitude and nature (ac or dc) specified.

A.3.3 Points of application. The test voltage shall be applied between mutually insulated portions of the specimen or between insulated portions and ground as specified. The method of connection of the test voltage to the specimen should be specified only when it is a significant factor.

A.3.4 Rate of application. The test voltage shall be raised from zero to the specified value as uniformly as possible, at a rate of approximately 500 volts (rms or dc) per second, unless otherwise specified. At the option of the manufacturer, the test voltage may be applied instantaneously during in-plant conformance testing.

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A.3.5 Duration of application. Unless otherwise specified, the test voltage shall be maintained at the specified value for a period of 60 seconds for qualification testing. For in-plant conformance testing, when specified, reduced time with a possible correlated higher test voltage may be used. Specimens with movable parts shall be tested as specified, in a manner to assure that repeated stresses are not applied to the same dielectric. Upon completion of the test, the test voltage shall be gradually reduced to avoid voltage surges. At the option of the manufacturer, the test voltage may be removed instantaneously during in-plant conformance testing.

A.3.6 Examination and measurement of specimen. During the dielectric withstanding voltage test, the fault indicator shall be monitored for evidence of disruptive discharge and leakage current. Following this, the specimen shall be examined and measurements shall be performed to determine the effect of the dielectric withstanding voltage test on specific operating characteristics, when specified.

A.3.7 Precautions. The dielectric withstanding voltage test should be used with caution particularly in in-plant conformance testing, as even an overpotential less than the breakdown voltage may injure the insulation and thereby reduce its safety factor. Therefore, repeated application of the test voltage on the same specimen is not recommended. In cases when subsequent application of the test potential is specified in the test routine, it is recommended that the succeeding tests be made at reduced potential. When either alternating-current (ac) or direct-current (dc) test voltages are used, care should be taken to be certain that the test voltage is free of recurring transients or high peaks. Direct potentials are considered less damaging than alternating potentials which are equivalent in ability to detect flaws in design and construction. However, the latter are usually specified because high alternating potentials are more readily obtainable. Suitable precautions must be taken to protect test personnel and apparatus because of the high potentials used.

A.3.8 Factors affecting use. Dielectric behavior of gases, oils, and solids is affected in various degrees by many factors, such as atmospheric temperatures, moisture, and pressure; condition and form of electrodes; frequency, waveform, rate of application, and duration of test voltage; geometry of the specimen; position of the specimen (particularly oil-filled components); mechanical stresses; and previous test history. Unless these factors are properly selected as required by the type of dielectric, or suitable correction factors can be applied, comparison of the results of individual dielectric withstanding voltage tests may be extremely difficult.

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APPENDIX A

A.4 APPARATUS

A.4.1 High-voltage source. The nature of the potential (ac or dc) shall be as specified. When an alternating potential is specified, the test voltage provided by the high-voltage source shall be nominally 60 hertz in frequency and shall approximate, as closely as possible, a true sine wave in form. Other commercial power frequencies may be used for in-plant conformance testing, when specified. All alternating potentials shall be expressed as root-mean-square values, unless otherwise specified. The kilovolt-ampere rating and impedance of the source shall be such as to permit operation at all testing loads without serious distortion of the waveform and without serious change in voltage for any setting. When the test specimen demands substantial test source power capacity, the regulation of the source shall be specified. When a minimum kilovolt-ampere rating is required, it shall be specified. When a direct potential is specified, the ripple content shall not exceed 5% rms of the test potential. When required, a suitable current-limiting device shall be used to limit current surges to the value specified.

A.4.2 Voltage measuring device. A voltmeter shall be used to measure the applied voltage to an accuracy of at least 5%, unless otherwise specified. When a transformer is used as a high-voltage source of alternating potential, a voltmeter connected across the primary side or across a tertiary winding may be used provided it is previously determined that the actual voltage across the test specimen will be within the allowable tolerance under any normal load condition.

A.4.3 Leakage current measuring device. When any leakage current requirements is specified, a suitable method shall be used to measure the leakage current to an accuracy of at least 5% of the specified requirement.

A.4.4 Fault indicator. Suitable means shall be provided to indicate the occurrence of disruptive discharge and leakage current in case it is not visually evident in the specimen. The voltage measuring device of A.4.2, the leakage current measuring device of A.4.3, or an appropriate indicator light or an overload protective device may be used for this purpose.

A.5 SUMMARY. The following details are to be specified in the individual specification:

a. Special preparations or conditions, if required (see A.3.1).
b. Magnitude of test voltage (see A.3.2).

    1. Test voltage, and duration for in-plant conformance testing, if different than for qualification testing (see A.3.5).

c. Nature of potential (ac or dc) (see A.4.1).

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d. Duration of application of test voltage for qualification testing if other than 60 seconds (see A.3.5).
e. Points of application of test voltage (see A.3.3).

    1. Method of testing specimens with movable parts (see A.3.5).

f. Method of connection of test voltage to specimen, if significant (see A.3.3).
g. Regulation, when applicable (see A.4.1).
h. Minimum kilovolt-ampere rating of high-voltage source, if required (see A.4.1).
i. Limiting value of surge current, if applicable (see A.4.1).
j. Maximum leakage current requirement, if applicable (see A.4.3).
k. Measurements after dielectric withstanding voltage test, if required (see A.3.6).

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APPENDIX B

INSULATION RESISTANCE

B.1 SCOPE

B.1.1 Scope. The appendix details the procedure for measuring the resistance offered by the insulating members of a component part to an impressed direct voltage tending to produce a leakage of current through or on the surface of these members. A knowledge of insulation resistances is important, even when the values are comparatively high, as these values may be limiting factors in the design of high-impedance circuits. Low insulation resistances, by permitting the flow of large leakage currents, can disturb the operation of circuits intended to be isolated, for example, by forming feedback loops. Excessive leakage currents can eventually lead to deterioration of the insulation by heating or by direct-current electrolysis. Insulation-resistance measurements should not be considered the equivalent of dielectric withstanding voltage or electric breakdown tests. A clean, dry insulation may have a high insulation resistance, and yet possess a mechanical fault that would cause failure in the dielectric withstanding voltage test. Conversely, a dirty, deteriorated insulation with a low insulation resistance might not break down under a high potential. Since insulating members composed of different materials or combinations of materials may have inherently different insulation resistances, the numerical value of measured insulation resistance cannot properly be taken as a direct measure of the degree of cleanliness or absence of deterioration. The test is especially helpful in determining the extent to which insulating properties are affected by deteriorative influences, such as heat, moisture, dirt, oxidation, or loss of volatile materials.

B.2 APPLICABLE DOCUMENTS

(This section is not applicable to this appendix.)

B.3 PROCEDURE

B.3.1 Preparation and measurement. When special preparations or conditions such as special test fixtures, reconnections, grounding, isolation, low atmospheric pressure, humidity, or immersion in water are required, they shall be specified. Insulation-resistance measurements shall be made between the mutually insulated points or between insulated points and ground, as specified. When electrification time is a factor, the insulation-resistance measurements shall be made immediately after a 2-minute period of uninterrupted test voltage application, unless otherwise specified. However, if the instrument-reading indicates that an insulation resistance meets the specified limit, and is steady or increasing, the test may be terminated before the end of the specified period. When more than one measurement is specified, subsequent measurements of insulation resistance shall be made using the same polarity as the initial measurements.

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APPENDIX B

B.3.2 Factors affecting use. Factors affecting insulation-resistance measurements include temperature, humidity, residual charges, charging currents of time constant of instrument and measured circuit, test voltage, previous conditioning, and duration of uninterrupted test voltage application (electrification time). In connection with this last-named factor, it is characteristic of certain components (for example, capacitors and cables) for the current to usually fall from an instantaneous high value to a steady lower value at a rate of decay which depends on such factors as test voltage, temperature, insulating materials, capacitance, and external circuit resistance. Consequently, the measured insulation resistance will increase for an appreciable time as test voltage is applied uninterruptedly. Because of this phenomenon, it may take many minutes to approach maximum insulation- resistance readings, but specifications usually require that readings be made after a specified time, such as 1 or 2 minutes. This shortens the testing time considerably while still permitting significant test results, provided the insulation resistance is reasonably close to steady-state value, the current versus time curve is known, or suitable correction factors are applied to these measurements. For certain components, a steady instrument reading may be obtained in a matter of seconds. When insulation-resistance measurements are made before and after a test, both measurements should be made under the same conditions.

B.4 APPARATUS. Insulation-resistance measurements shall be made on an apparatus suitable for the characteristics of the component to be measured such as a megohm bridge, megohm-meter, insulation-resistance test set, or other suitable apparatus. Unless otherwise specified, the direct potential applied to the specimen shall be 500 volts $\pm$ 10%. For in-plant conformance testing, any voltage may be used provided it is equal to or greater than the minimum potential allowed by the applicable test condition. Unless otherwise specified, the measurement error at the insulation-resistance value required shall not exceed 10%. Proper guarding techniques shall be used to prevent erroneous readings due to leakage along undesired paths.

B.5 SUMMARY. The following details are to be specified in the individual specification:

a. Test condition letter, or other test potential, if specified (see B.4).
b. Special preparations or conditions, if required (see B.3.1).
c. Points of measurement (see B.3.1).
d. Electrification time, if other than 2 minutes (see B.3.1).
e. Measurement error at the insulation-resistance value required, if other than 10% (see B.4).

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I RECOMMEND A CHANGE:	**1. DOCUMENT NUMBER** MIL-PRF-62331A(AT)	**2. DOCUMENT DATE *(YYMMDD)*** 970815

3. DOCUMENT TITLE
Pump Assembly, Bilge, Electric Motor Driven, Direct Current

4. NATURE OF CHANGE ***(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)***

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*	**b. ORGANIZATION**	
c. ADDRESS *(Include Zip Code)*	**d. TELEPHONE *(Include Area Code)*** **(1) Commercial** **(2) AUTOVON** ***(If applicable)***	**7. DATE SUBMITTED *(YYMMDD)***

8. PREPARING ACTIVITY

a. NAME	**b. TELEPHONE *(Include Area Code)*** **(1) Commercial** (810) 574-8745 **(2) AUTOVON** 786-8745
c. ADDRESS *(Include Zip Code)* **Commander** **U.S. Army Tank-automotive and Armaments Command** **ATTN: AMSTA-TR-E/BLUE** **Warren, MI 48397-5000**	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** **Defense Quality and Standardization Office** **5203 Leesburg Pike, Suite 1403** **Falls Church, VA 22041-3466** **Telephone (703) 756-2340 AUTOVON 289-2340**

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